Harvard Architecture

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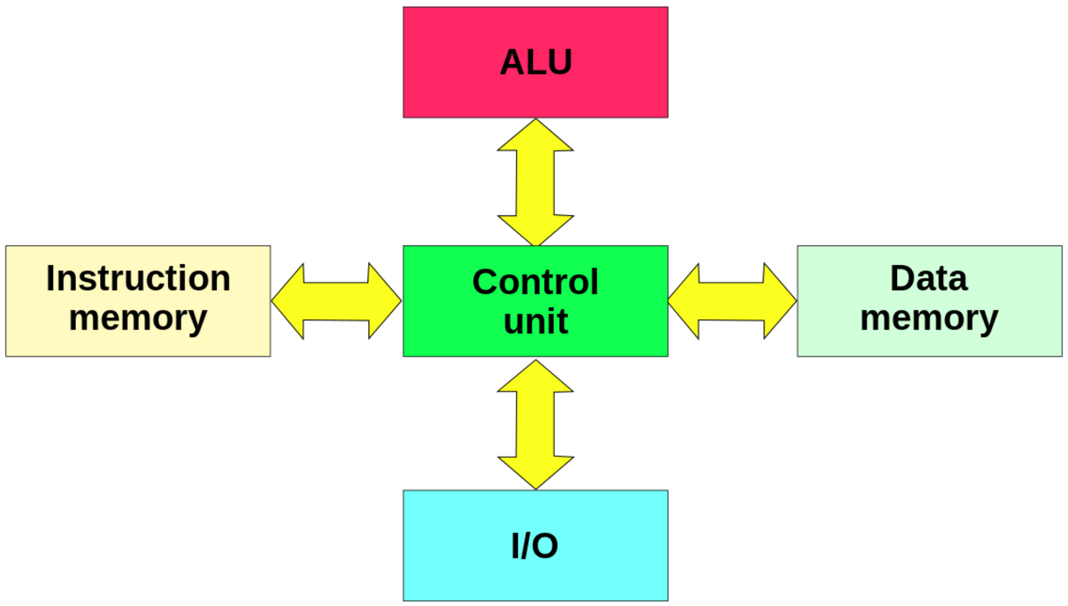
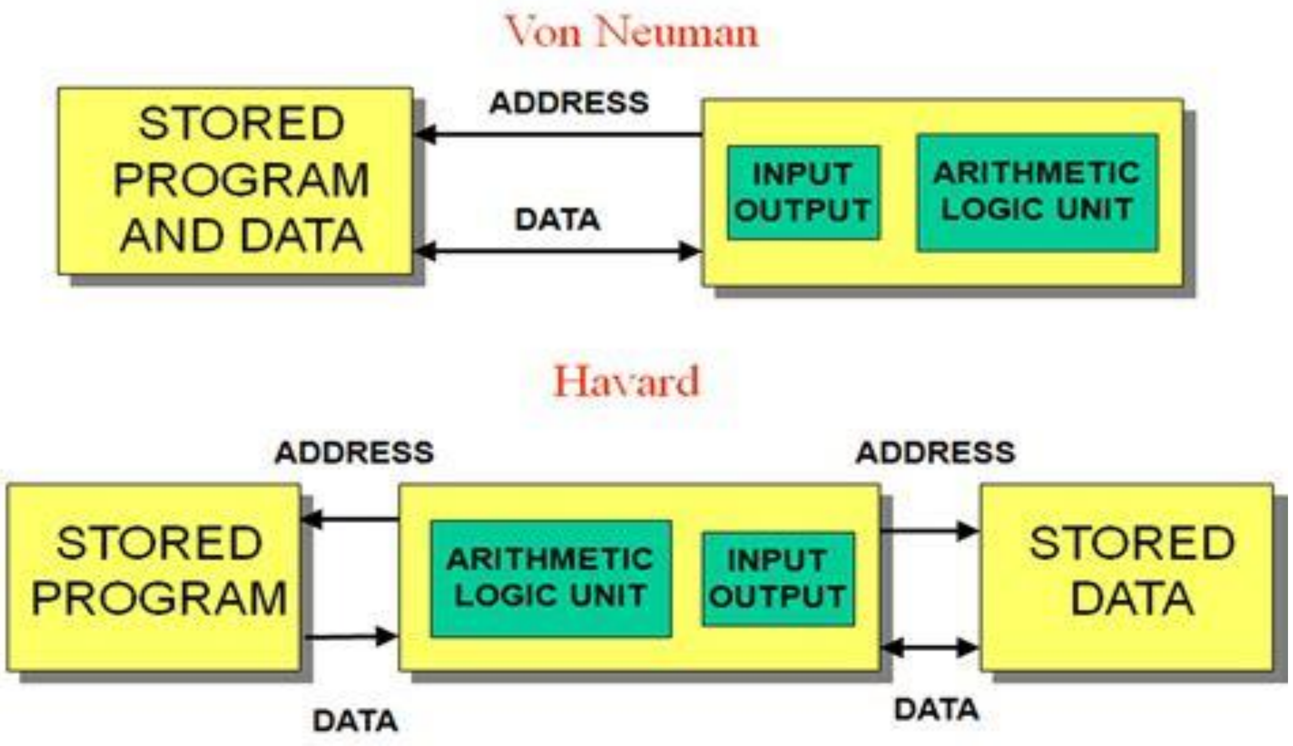
Harvard Architecture



Harvard Architecture

- The **Harvard architecture** is a computer architecture with physically separate storage and signal pathways for instructions and data.
- It was basically developed to overcome the bottleneck of Von Neumann Architecture.
- The main advantage of having separate buses for instruction and data is that CPU can access instructions and read/write data at the same time.
- Programs needed to be loaded by an operator; the processor could not initialize itself.
- In a Harvard architecture, there is no need to make the two memories share characteristics.
- In particular, the word width, timing, implementation technology, and memory address structure can differ.
- Instructions – read-only memory
- Data memory – read-write memory.
- In some systems, there is much more instruction memory than data memory so instruction addresses are wider than data addresses.

Harvard Vs Von-Neumann Architecture



Von-Neumann Vs Harvard

- Von-Neumann Architecture
- The data and program are stored in the same memory
- The code is executed serially and takes more clock cycles.
- The programs can be optimized in lesser size.
- Parallel access to data and program is not possible.
- One bus is simpler for the control unit design.
- Error in program can rewrite instruction and crash program execution

- Harvard Architecture
- The data and program memory are separate. Both memories can use different sizes.
- The code is executed in parallel.
- The program tend to grow big in size.
- Two memories with two buses allow parallel access to data access and instructions.
- Control unit for 2 buses is more complicated and more expensive.
- Program can't write itself

Modified Harvard Architecture

- A modified Harvard architecture machine is very much like a Harvard architecture machine, but it relaxes the strict separation between instruction and data while still letting the CPU concurrently access two (or more) memory buses.
- The most common modification includes separate instruction and data caches backed by a common address space.
- While the CPU executes from cache, it acts as a pure Harvard machine.
- When accessing backing memory, it acts like a von Neumann machine

Modified Harvard Architecture